

Response to Reviewer Comments

Response to Second-Round Reviewer Comments

We thank the reviewer for the detailed and constructive second-round feedback (6/10). Below we address each of the five new concerns.

New W1: Generalization scope still too broad

Concern. The paper’s claims still read as general conclusions about latent world models.

Response. We have further narrowed scope language. The abstract now reads “For LeWM-style latent planners...” and the Cube $m = 64$ success has been corrected from 72.0% to 56.0% to match the extended 50-pair Table 2 data. We position the paper as a rigorous LeWM-specific audit that proposes a reusable diagnostic protocol, not a universal empirical law.

New W2: V1 oracle cannot alone complete mechanism attribution

Concern. V1 uses physical variables directly. It cannot distinguish learned representation failure from predictor rollout error or cost shape issues. A V3 (actual-terminal encoder latent cost) comparison on the same pool is needed.

Response. We conducted the requested V3 same-pool ranker decomposition on all 100 PushT final pools. For each of the 300 candidates per pair, we replayed the raw actions in the simulator, encoded the actual terminal observation with the LeWM encoder, and computed $C_{V3} = \|z_{\text{terminal}} - z_g\|_2^2$.

Cost ranker	R_{pool} (mean $\pm$ std)	95% bootstrap CI
C_{model} (predicted terminal latent)	0.084 ± 0.243	[0.037, 0.131]
C_{V3} (actual terminal latent)	0.258 ± 0.566	[0.146, 0.365]
C_{V1} (physical hinge oracle)	0.666 ± 0.357	[0.595, 0.735]

The decomposition reveals a three-level attribution. Removing predictor error ($C_{\text{model}} \rightarrow C_{V3}$) improves R_{pool} by 0.174, confirming that predictor rollout degrades pool ranking. Removing both predictor and encoder geometry ($C_{V3} \rightarrow C_{V1}$) improves R_{pool} by a further 0.408. Even with perfect prediction, the encoder’s Euclidean latent distance remains a weak pool ranker.

This directly addresses the reviewer’s concern: the problem is not solely predictor error, and the encoder’s terminal-latent geometry / Euclidean cost shape is the dominant bottleneck. We have revised Section 4.4 to include this decomposition table.

Location. Section 4.4, new decomposition table and revised interpretation.

New W3: Statistical uncertainty insufficient

Concern. Key claims like “22% CEM-specific” and “78% representation-driven” lack confidence intervals.

Response. We computed bootstrap 95% CIs (10,000 resamples) for all core metrics:

- $\Delta_{\text{CEM}} = 0.422$ with CI $[0.374, 0.468]$.
- $R_{\text{pool}}(C_{\text{model}}) = 0.084$ with CI $[0.037, 0.131]$.
- $R_{\text{pool}}(C_{V1}) = 0.666$ with CI $[0.595, 0.735]$.
- MPPI attribution = 22% with CI $[-16\%, 48\%]$; the point estimate is positive but the CI spans zero.

We have revised the MPPI discussion to present the attribution as a point estimate with acknowledged uncertainty rather than a precise decomposition. The qualitative finding – that soft weighting improves but does not eliminate the endpoint-pool gap – is robust regardless of the exact percentage.

For repair failures, we added Clopper–Pearson exact binomial CIs (e.g., cost head: 0/16, 95% upper bound = 20.6%).

Location. Core metrics in Sections 4.4, 7, and 8 now include bootstrap CIs or exact intervals where applicable.

New W4: Cube is Case B, not planning failure

Concern. Cube has $R_{\text{pool}} \approx 0$ but high success, fitting Case B (pool already good, ranking less important) rather than planning failure.

Response. We agree and have added an explicit sentence in Section 4.2: “In Cube, the near-zero R_{pool} coexists with relatively high planning success, consistent with Case B in our taxonomy: the pool contains sufficient success mass that ranking is less critical for action selection. Cube therefore provides cross-environment evidence for the endpoint-pool ranking gap, not for cost-driven planning failure.”

Location. Section 4.2, after Table 2 discussion.

New W5: Text consistency (72.0% vs 56.0%, missing Section 7)

Concern. Abstract Cube number conflicts with Table 2; Section 7 is missing.

Response. Fixed. The abstract now reports 56.0% matching the extended 50-pair Table 2 data. The section numbering gap has been removed (now flows Section 6 → Section 7 Discussion → Section 8 Limitations). The 72.0% figure from the original 25-pair single-seed run remains only in the appendix seed-level sanity table, clearly labeled as preliminary.

Location. Abstract, section headers throughout.

Response to First-Round Reviewer Comments

We thank the reviewer for the thorough and constructive evaluation. Below we address each weakness with new experiments and text revisions. All supplementary experiments, scripts, and data are released in the repository under `scripts/revision/`, `results/revision/`, and `docs/revision/`.

W1: Single-model, dual-environment generalization

Reviewer concern. All experiments use only LeWM on PushT and OGBench-Cube. The abstract and introduction overclaim generality with phrases like “any latent world model that plans with terminal latent costs can be audited this way.”

Our response. We agree that the empirical evidence is bounded to one model and two environments, and have revised all overclaiming language accordingly.

Abstract: The original sentence has been replaced with: “For LeWM-style latent planners that select actions by terminal latent costs under iterative sampling-based optimization, we recommend reporting pool-level and selection-level metrics alongside endpoint metrics.”

Introduction (Section 1): We now state explicitly: “Our empirical claims apply to LeWM with Euclidean terminal latent costs under CEM-style optimization, evaluated on PushT and OGBench-Cube. The evaluation principle is broader, but it should be applied with the same instrumentation.”

Limitations (Section 8): Three explicit scope boundaries are now listed: (1) single model, two environments; (2) CEM as primary optimizer with partial MPPI cross-check; (3) privileged physical cost required for full evaluation.

Cross-model feasibility: We investigated PLDM (Sobal et al., 2025) as a cross-model validation target. The public repository exists but no ready-to-use PushT checkpoint was available at revision time. We leave cross-model validation as future work, noting that the diagnostic protocol (R_{endpoint} , R_{pool} , Δ_{CEM}) is model-agnostic and directly applicable to any latent world model that exposes its candidate pool.

Location in revised paper. Abstract (page 1), Section 1 (page 2, lines 88–97), Section 8 (page 12).

W2: Diagnostic power of Δ_{CEM} – attribution of $R_{\text{pool}} \approx 0$

Reviewer concern. $R_{\text{pool}} \approx 0$ could reflect multiple mechanisms: (a) CEM convergence compresses candidates into a physically narrow neighborhood where any cost fails to discriminate, (b) the encoder genuinely loses local order preservation, or (c) predictor bias. The paper does not experimentally distinguish these.

Our response. We conducted a new $R_{\text{pool}}(\text{V1})$ attribution experiment on all 100 PushT default CEM final pools. For each pair, we computed the Spearman correlation of the V1 oracle hinge cost (a privileged physical cost) against $C_{\text{real_state}}$ within the same 300-candidate pool, and compared it with the learned Euclidean cost.

Key results:

Subset	n	$R_{\text{pool}}(C_{\text{model}})$	$R_{\text{pool}}(C_{V1})$
Overall	100	0.084	0.666
Invisible quadrant	16	0.022	0.775
Sign reversal	21	-0.040	0.925
Latent-favorable	12	0.197	0.417
Ordinary	47	0.146	0.581

The V1 oracle retains strong pool-level ranking ($R_{\text{pool}}(C_{V1}) = 0.666$) where the learned cost fails ($R_{\text{pool}}(C_{\text{model}}) = 0.084$). This rules out pure CEM convergence compression as the dominant explanation: if the pool were compressed into a physically indistinguishable neighborhood, V1 would also fail. Instead, the data clearly indicates **local representation failure**: the learned Euclidean cost loses order-preservation in the CEM-converged region while the privileged oracle cost does not.

The effect is strongest in the invisible quadrant (16 pairs where endpoint metrics look healthy but planning fails): $R_{\text{pool}}(C_{V1}) = 0.775$ while $R_{\text{pool}}(C_{\text{model}}) = 0.022$, with pool success mass of only 1.4%. The CEM pool contains successful candidates, but the learned cost cannot identify them.

Location in revised paper. New Section 4.4, “Mechanism attribution: convergence compression versus local representation failure” (page 8).

W3: Weak Cube evidence

Reviewer concern. Cube full projected CEM uses only 25 pairs $\times$ 1 seed (vs. 100 pairs $\times$ 3 seeds for re-rank-only). The inverted-U pattern received MIXED status. This asymmetry weakens cross-environment claims.

Our response. We extended the Cube full projected CEM experiment from 25 pairs $\times$ 1 seed to **50 pairs $\times$ 3 projection seeds** across all five projection dimensions ($m \in \{1, 8, 32, 64, 192\}$), totaling 750 CEM rollouts with full 300-candidate simulator scoring (~ 10 hours on Apple Silicon MPS).

Updated results:

m	Full-CEM success (50p, 3s)	Re-rank success (100p, 3s)	Gap
1	32.7% $\pm$ 8.1%	42.7% $\pm$ 3.2%	-10.0 pp
8	52.7% $\pm$ 4.2%	43.3% $\pm$ 1.5%	+9.3 pp
32	61.3% $\pm$ 1.2%	46.3% $\pm$ 2.5%	+15.0 pp
64	56.0% $\pm$ 5.3%	49.7% $\pm$ 2.5%	+6.3 pp
192	62.0% $\pm$ 6.0%	47.7% $\pm$ 1.2%	+14.3 pp

$R_{\text{pool}}(C_{\text{model}})$ across all dimensions ranges from -0.001 to 0.023 , consistently near zero, confirming endpoint-planning decoupling in Cube at scale.

The original inverted-U success pattern (peak at $m = 32$, decline at $m = 192$) **does not persist** at 50-pair, 3-seed scale. Success plateaus from $m = 32$ onward, indicating the original pattern was small-sample noise. This simplifies the cross-environment comparison: both PushT and Cube show rising R_{endpoint} with dimension while R_{pool} remains near zero.

Table 2 in the revised paper now uses the extended 50-pair $\times$ 3-seed Cube data.

Location in revised paper. Table 2 (page 7), Section 4.2 discussion (page 7, lines 368–377), Section 8 Limitations (page 12).

W4: No alternative optimizer comparison

Reviewer concern. The paper does not test whether endpoint-planning decoupling is CEM-specific (due to hard elite truncation) or generalizes to other iterative optimizers like MPPI.

Our response. We implemented an MPPI planner that replaces CEM’s hard elite selection with softmax-weighted averaging over all 300 candidates, keeping all other parameters identical (same model, action parameterization, horizon, 30 iterations, 300 candidates). After a temperature sweep ($\tau \in \{0.01, 0.1, 1.0, 10.0\}$), we selected $\tau^* = 1.0$ and ran the experiment on 30 PushT pairs $\times$ 3 seeds (90 scored pools).

Comparison:

Metric	CEM default	MPPI ($\tau = 1.0$)
Planning success	53.3%	38.9%
$R_{\text{pool}}(C_{\text{model}})$	0.115	0.194
$R_{\text{pool}}(C_{V1})$	0.537	0.707
Pool $C_{\text{real_state}}$ std	7.947	29.478

MPPI preserves **3.7 $\times$ more physical diversity** in its final pool than CEM (std 29.478 vs. 7.947), confirming that CEM’s hard truncation compresses pool diversity. Correspondingly, $R_{\text{pool}}(C_{\text{model}})$ improves from 0.115 to 0.194 under MPPI.

However, MPPI still retains only $\sim 39\%$ of the endpoint ranking signal ($R_{\text{pool}}/R_{\text{endpoint}} = 0.194/0.495$). Using $R_{\text{endpoint}} = 0.495$ at $m = 192$ as the reference, the CEM-to-MPPI improvement has a 22% point estimate of the endpoint-pool gap, but the paired-bootstrap 95% confidence interval is wide ($[-16\%, 48\%]$). Thus endpoint-planning decoupling is **partially optimizer-specific** (CEM’s hard truncation worsens pool geometry) and **partially optimizer-general**: the majority of the decoupling appears representation-driven, with the MPPI improvement modest relative to the full endpoint-pool gap.

The lower MPPI planning success (38.9% vs. 53.3%) is consistent with slower convergence under 30 iterations and does not invalidate the pool-level comparison, which is the relevant diagnostic.

Location in revised paper. New Section 7.3, “Optimizer dependence: CEM versus MPPI” (page 12).

W5: Privileged physical cost requirement limits practical applicability

Reviewer concern. All pool-level metrics require $C_{\text{real_state}}$ (simulator access). How can one detect Δ_{CEM} in deployment without a simulator?

Our response. We tested whether privilege-free proxy metrics (computable from the learned cost alone, no simulator needed) correlate with selection regret across the 100 PushT pairs.

Results:

Proxy metric	Corr. with selection regret	<i>p</i> -value
top30_Cmodel_std	0.314	0.001
pool_Cmodel_std	0.254	0.011
C_model_dynamic_range	0.223	0.026

Within the invisible-quadrant subset, the signal is stronger: `pool_Cmodel_std` vs. selection regret has $\rho = 0.588$ ($p = 0.017$). As a cross-check, `pool_Cmodel_std` tracks `pool_Creal_std` at $\rho = 0.482$ ($p < 0.001$), confirming that learned-cost spread partially reflects physical spread.

No single global proxy crosses our pre-specified $\rho > 0.4$ threshold, so these proxies are not full replacements for the privileged audit. However, they demonstrate that **partial monitoring is feasible without simulator access**: low learned-cost diversity in the CEM final pool can flag candidate-pool compression and elevated selection risk.

Location in revised paper. Section 7.1, paragraph on privilege-free monitoring (page 11, lines 584–593).

Summary of Changes

Weakness	Experiment	Key finding	Paper location
W1	Text revision + PLDM feasibility	Scope narrowed; PLDM deferred	Abstract, Sec. 1, Sec. 8
W2	Rpool(V1) attribution (100 pairs)	Local representation failure dominant	New Sec. 4.4
W3	Cube extended (50p $\times$ 3s, 750 runs)	Decoupling robust at scale; inverted-U gone	Table 2, Sec. 4.2
W4	MPPI comparison (30p $\times$ 3s, 90 pools)	Soft weighting improves but does not close the gap; attribution CI is wide	New Sec. 7.3
W5	Privilege-free proxy (100 pairs)	Partial monitoring feasible (ρ up to 0.588)	Sec. 7.1

All supplementary scripts, results, and memos are available in the repository under `scripts/revision/`, `results/revision/`, and `docs/revision/`. The progress document `docs/revision/revision_progress.md` provides a complete timeline and artifact inventory.